# Abhishek Kumar

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## PROFESSIONAL SUMMARY

Computer Science & Engineering graduate (2026) skilled in Python backend development with FastAPI, PostgreSQL, and SQLAlchemy. Built InkFlow, a full-stack blog platform with JWT authentication and Docker/Nginx deployment, and ByteSniffer, a Python network traffic analyzer. Seeking an entry-level Software Engineer / Python Backend Developer role.

## TECHNICAL SKILLS

**Languages:** Python, SQL, JavaScript

**Backend & APIs:** FastAPI, RESTful API Design, Swagger/OpenAPI, JWT Authentication, SQLAlchemy ORM, Alembic, Pydantic, Node.js/Express (basic)

**Databases:** PostgreSQL, MySQL, SQLite, MongoDB

**Tools & DevOps:** Git, GitHub, Docker, Nginx, AWS S3, Postman, VS Code

**Testing:** Pytest, Automation Testing (KodNest internship)

**Frontend (Basic):** HTML, CSS, Bootstrap, Jinja2, React.js

**Concepts:** Object-Oriented Programming, Networking Fundamentals (TCP/IP, TLS), Debugging

## PROJECTS

**InkFlow** | REST API & Full-Stack Web Development | GitHub

***Technologies:*** *Python, FastAPI, SQLAlchemy, PostgreSQL, Pydantic, JWT, Jinja2, JavaScript, Bootstrap*

- Developed a full-stack blog platform using FastAPI, exposing RESTful APIs to create, retrieve, update, and delete blog posts
- Integrated PostgreSQL with SQLAlchemy ORM to manage user accounts, blog posts, and relational data
- Implemented user registration, login, JWT-based authentication, password hashing, and authorization for protected endpoints
- Built a responsive web interface with Jinja2 templates, Bootstrap, and JavaScript that consumes the backend APIs
- Added input validation, exception handling, pagination, and profile image uploads to improve reliability and usability
- Structured the backend into modular API routers and managed database schema migrations with Alembic
- **Testing & Deployment:** Pytest for testing; deployed using Docker, AWS S3, Nginx, and SSL

**ByteSniffer** | Deep Packet Inspection & Network Traffic Analysis | GitHub

***Technologies:*** *Python, TCP/IP, TLS, PCAP*

- Built a Python-based Deep Packet Inspection tool that reads PCAP captures, parses Ethernet, IPv4, TCP, and UDP headers, and tracks connections using 5-tuple flow identification
- Implemented TLS SNI and HTTP Host header extraction to classify applications (YouTube, Facebook, Google) even in encrypted HTTPS traffic
- Designed a rule-based blocking system for IP, application, and domain rules, enforced at the flow level, with filtered traffic written to a new PCAP
- Generated per-application traffic reports (forwarded vs. dropped packets) and created Python scripts to produce test PCAP data

## EXPERIENCE

**Automation Testing Intern — KodNest (Online)** *Feb 2026 – May 2026*

- Applied automation testing concepts, tools, and workflows to validate application functionality and identify defects
- Practiced structured testing and debugging, building quality-assurance skills that complement backend and API development

## EDUCATION

**Sir M. Visvesvaraya Institute of Technology** *2022 – 2026*

Bachelor of Engineering in Computer Science & Engineering | CGPA: 8.24 | Bengaluru, Karnataka

## ACHIEVEMENTS

**Hack-Nocturne (NextKick) 2025 | Contributor** *March 2025*

- Contributed to an AI-based football scouting platform built with Node.js, Prisma, PostgreSQL, and YOLOv8